Science of the Total Environment 813 (2022) 152082

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Mechanisms of grazing management impact on preferential water flow and infiltration patterns in a semi-arid grassland in northern China

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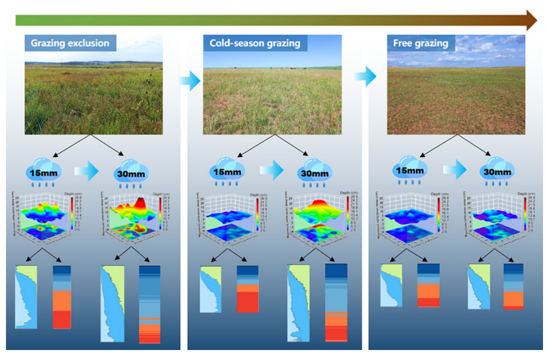
HIGHLIGHTS

- Grazing management affects soil and vegetation properties and root traits.
- Long-term grazing exclusion promotes preferential water infiltration into deep soil.
- Variations of preferential flow can be explained by the changes in pore systems.
- Infiltration amount affects the macropore flow–soil matrix interaction.
- New evidence of the eco-hydrological process driven by grazing management.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 7 August 2021
Received in revised form 12 November 2021
Accepted 26 November 2021
Available online 30 November 2021

Editor: Jurgen Mahlknecht

Keywords:
Grazing management
Preferential flow
Dye tracer
Grassland degradation
Semi-arid area

ABSTRACT

Grazing management is widely used to control grassland degradation in Inner Mongolia. However, the correlation between the soil physical properties, root traits, and infiltration patterns of different types of grazing management has seldom been studied. To reveal the effect of grazing management on water infiltration and preferential flow behavior, we first investigated the soil and plant properties in a grazing exclusion (19 years, GE), cold-season grazing (19 years, CG), and free-grazing grassland (19 years, FG) in a semi-arid grassland in Inner Mongolia. Dye tracer infiltration was adopted to obtain the water infiltration patterns from different types of grazing management. Finally, root biomass and root morphological traits were measured in a field experiment. The results showed that the plant height, vegetation coverage, richness index, Shannon-Wiener index, soil water content, total porosity, and mean weight diameter were higher at the GE site than at the FG site, whereas soil bulk density and sand content were lower at the GE site than at the FG site ($P < 0.05$). In addition, the root mean diameter, specific root length, and root mass density were higher at the GE site than at the FG site. As a result, differences in these root traits and soil and vegetation properties affected the preferential water flow behavior in the three types of grassland. The preferential flow evaluation index (P_{FI}) of the GE, CG, and FG sites was 0.89, 0.30, and 0.15, respectively, which indicated that more obvious preferential flow occurred at the GE site than at the CG and FG sites. These findings highlight that the long-term GE enhanced plant density and root biomass, which could potentially promote the natural restoration of soil pores and preferential water infiltration. Therefore, local governments and herders should implement GE rather than other grazing management practices to prevent grassland degradation.

1. Introduction

Grasslands occupy approximately 40% of the Earth's surface area, amounting to 50 million km^2, and provide a series of important services and functions (Du and Gao, 2021; Tian et al., 2021). Grazing is a widespread land-use practice in grassland ecosystems, and it exerts a profound

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http://dx.doi.org/10.1016/j.scitotenv.2021.152082

effect on ecosystem services and functions by changing the composition of plant species and soil processes (Talore et al., 2015; Sigcha et al., 2018). However, many grasslands are facing the threat of long-term overgrazing, which may lead to a significant decrease in canopy coverage, productivity, plant species diversity, and soil porosity (Wang et al., 2016; Byrnes et al., 2018; Yu et al., 2020). Desert steppes in Inner Mongolia are sensitive to climate change and human disturbance, and long-term overgrazing has caused large areas of soil and vegetation degradation and even grassland desertification (Zhao et al., 2018; Wu et al., 2021). Ecological restoration, associated with grazing exclusion or a reduction in grazing intensity, has been implemented in recent decades to control grassland degradation in Inner Mongolia (Wang et al., 2017).

The key factor to ecological restoration in arid and semi-arid ecosystems is to maintain soil water storage (Zhu et al., 2020). Limited rainfall amounts and high rates of potential evapotranspiration lead to low water storage in the vadose zone in arid and semi-arid areas (Zhang et al., 2018a). Preferential infiltration of precipitation is an important source of groundwater recharge in semi-arid grasslands. It is quite significant since soil infiltration is the limiting factor for runoff generation and vegetation development (Mei et al., 2018). Preferential flow, which bypasses the soil matrix area through the soil macropores, allowing rapid penetration to greater depths, plays a critical role in the hydrological process (Edith Allaire et al., 2011; Bargués Tobella et al., 2014). Although the preferential flow pathway accounts for a limited proportion of the total pore volume of the soil, the contribution of preferential flow to water infiltration may reach 15%–85% (Laine Kaulio et al., 2014; Zhang et al., 2018b; Zhang et al., 2021). Preferential flow can significantly affect the degradation of arid areas by affecting the infiltration process, reducing runoff and soil erosion, and increasing root zone water replenishment (Wang et al., 2020; Zhu et al., 2020). Therefore, it is critical to assess the level of preferential infiltration because of its role in the ecological functions of the soil (Jiang et al., 2019), including surface runoff (Chaplot and Poesen, 2012), soil degradation (Jiang et al., 2021), soil water storage (Zhang et al., 2018a), and groundwater pollution (Jørgensen et al., 2002; Kramers et al., 2009).

The dye tracer method is widely used as a relatively cheap and fast method to characterize infiltration patterns, especially for identifying preferential flow types in soil profiles (Yi et al., 2020; Liu et al., 2021). According to Weiler and Flühler (2004), preferential water flow can be classified as macropore flow with low interaction, mixed interaction, and high interaction, depending on the interactions between macropores and the soil matrix. Several studies have demonstrated that the spatial and temporal variations of soil preferential flow are primarily influenced by complex interactions such as topographic features, soil properties, vegetation types, and faunal activities (Van Schaik, 2009; Liu and Lin, 2015; Maier et al., 2020; Liu et al., 2021). Highly heterogeneous soil architecture can strongly affect the generation of preferential flow (Wang et al., 2018). Generally, soils with a higher sand content have high preferential flow (Mei et al., 2018), whereas soils with higher silt and clay contents are more likely to experience matrix flow (Maier et al., 2020). Recent studies highlight the influence of plant roots on preferential flow; the macropores formed by living and dead plant root channels can form preferential flow pathways (Cui et al., 2019; Guo et al., 2019; Luo et al., 2019; Wu et al., 2020). In semi-arid savanna rangeland areas, macropore pathways are most abundant under larger shrub canopies and less abundant under smaller shrub canopies (Marquart et al., 2020). In Burkina Faso (West Africa), scattered trees associated with large open areas in semi-arid landscapes enhanced soil infiltration and preferential flow (Bargués Tobella et al., 2014). On the Loess Plateau, North China, vegetation patchiness enhanced spatial variations of water infiltration and preferential flow, which resulted from increasing length and volume densities of fine roots (Wang et al., 2020).

The disturbance caused by soil mechanics, interlinked with hydrological changes, often adversely affects soil properties (Zhao et al., 2011). The dynamic stress changes in the soil under repeated loads caused by grazing associated with livestock activities cause soil deformation (Warren et al., 1986). In the topsoil especially, the characteristics of soil deformation are changes in pore size distribution and connectivity, which have a significant impact on the hydrological process (Hao et al., 2020a). Moreover, grazing activities significantly affect above- and below-ground vegetation characteristics and may cause changes in root morphology traits including root length, root diameter, and specific root length (Deng et al., 2014). Macropore channels formed by plant roots play a critical role in the infiltration process in arid and semi-arid regions. The dynamics of grazing pressure could further impact these processes under different types of grazing management. However, previous studies mainly focused on the influence of grazing management on soil and vegetation properties (Armitage et al., 2012; Yu et al., 2020; Du and Gao, 2021), and it is not clear how soil properties and root morphological traits are affected by grazing management and how these affect the preferential flow in the soil profile. The soil water content and soil water storage were significantly increased after 11 years of grazing exclusion, although the reason was not clear (Dong et al., 2015). Therefore, studies on preferential flow behavior and infiltration patterns in semi-arid grasslands are still needed, especially when including the effects of grazing management.

The present study was planned to (1) assess the effects of grazing management on soil properties, vegetation characteristics, and root morphology traits, (2) determine the vertical infiltration patterns and preferential flow behavior among three grazing management sites under 15/30 mm infiltration using the dye tracer and multi-index evaluation methods, and (3) identify the mechanisms through which grazing management impacts the preferential water flow and infiltration patterns in a semi-arid grassland in northern China.

2. Material and methods

2.1. Study area

This study was carried out in the Xilamuren Grassland (41°20′55″–41°21′23″N, 111°12′19″–111°12′43″E; 1600 m) of Inner Mongolia, North China. The area falls within a semi-arid continental climate, with an average temperature of 2.5 °C. The average annual precipitation is 282.4 mm, mainly concentrated from June to September (Meng et al., 2018). The major soil type is a Haplic kastanozem, with mean clay, silt, and sand contents of 4.48%, 16.81%, and 78.68%, respectively. The dominant natural plant species are *Stipa kryloyii*, *Aneurolepidium chinense*, *Stipa breviflora*, *Agropyron cristatum*, and *Artemisia frigida*.

2.2. Experimental design and field measurements

2.2.1. Experimental design

In the present study, three sites were chosen to represent different types of grazing management treatments: grazing exclusion (GE), cold-season grazing (CG), and free grazing (FG). The topographic and climatic conditions of the three study sites were similar, and the grasslands have a long history of livestock grazing from the 1960s, according to information obtained from herders. Since April 2002, 2.1-m-tall metal fences have been used to keep livestock away from GE areas. The CG areas were reserved for livestock grazing only during the cold season (from November to April), but during the other months, these areas were fenced with 2.1-m-tall metal fences since April 2002. The FG areas were managed under a free-grazing regime. The plot sizes of the GE, CG, and FG sites were 400 m × 300 m, 300 m × 250 m, and 400 m × 200 m, respectively. The grazing intensity was determined by the herbage allowance, which is a relationship between herbage and livestock (Schönbach et al., 2011). The grazing intensity of the CG site was 0.5–1.0 sheep ha^{-1} from November to April and no grazing activity from May to October. The grazing intensity of the FG site was 0.5–1.0 sheep ha^{-1} from November to April and 1.5–2.0 sheep ha^{-1} from May to October. The grazing intensity remained highest at the FG site, intermediate at the CG site, and minimal at the GE site. The three demonstration sites are unique and provide valuable information to study the long-term effects of grazing management on the grassland hydrological process. Similar to experiments along chronosequences, this long-term grazing management experiment was inevitably a pseudoreplicated experiment, but within this constraint, we have done our best to follow the methods used in chronosequences and long-term pseudoreplicated experimental studies (Hafner et al., 2012; Shi et al., 2013; Guilherme Pereira et al., 2019; Yu et al., 2020).

2.2.2. Plant and soil sampling

Plant and soil sampling was conducted simultaneously at the three sites from August 5 to 10 in 2020, and there were no precipitation events that occurred before or during sampling. We randomly selected three plots (20 m × 20 m) at each study site, and five 1 × 1 m^2 quadrats were randomly located within each plot. Any two quadrats were separated by at least 5 m. All of the plant species within each quadrat were identified, and their coverage, number, height, and frequency were recorded and then averaged as the species richness at the plot level (Table 1). The aboveground biomass (AGB) was determined by clipping the aboveground standing green parts. Soil samples were collected from the 0 to 40-cm soil layers in the four corners of each quadrat. Four 100-cm^3 soil cylinders (width × height of 5 cm × 5 cm) were used to measure the soil water content (SWC,%), bulk density (BD, g cm^{-3}), and soil total porosity (TP, %), while four 500-cm^3 soil samples were collected to determine the soil organic carbon (SOC, g kg^{-1}), aggregate stability, and soil texture (Table 2).

2.2.3. Plant and soil measurements

The plant species richness (R), evenness index (E), and Shannon-Wiener diversity index (H) of the GE, CG, and FG grasslands were calculated as follows (Liu et al., 2019):

Plant species richness (R):

$$R = S. \tag{1}$$

Evenness index (E):

$$E = \frac{H}{\ln S}. \tag{2}$$

Shannon-Wiener diversity index (H):

$$H = -\sum_{i=1}^{S} Pi \ln Pi, \tag{3}$$

where S is the total number of plant species in the sampling quadrat, and Pi is the weight proportion of the ith species.

The soil samples were labeled and air-dried after collection, and then, the SOC was analyzed via the dichromate oxidation method (Nelson and Sommers, 1983). The soil particle size distributions were measured by a Malvern Mastersizer-3000 (Malvern Instruments Ltd., Malvern, UK) in the laboratory. SWC was measured by oven-drying at 105 °C for 24 h to a constant weight (W_d). The soil aggregate stability was measured by a fast-wetting treatment (Hao et al., 2020b) and determined by the mean weight diameter (MWD, mm). BD, TP, and MWD were then calculated with the following formulae:

$$\mathrm{BD} = \frac{W_d - W_c}{V}, \tag{4}$$

$$\mathrm{TP} = \left(1 - \frac{\mathrm{BD}}{ds}\right) \times 100, \tag{5}$$

$$\mathrm{MWD} = \sum_{1}^{n+1} \frac{\mathrm{r_{i+1}} + \mathrm{r_i}}{2} \times \mathrm{m_i}, \tag{6}$$

where V is the soil cylinder volume, W_c is the cylinder weight (g), ds is the soil particle density (2.65 g cm^{-3} in this study), r_i is the aperture of the ith mesh (mm), m_i is the sediment weight fraction of the ith sieve, and n is the number of sieves.

2.3. Dye tracer experiment

The dye tracer experiment was carried out at the same time as the plant and soil sampling on August 5, 2020. A total of 18 (three sites × three plots × two infiltration amounts) dye tracer experiments were carried out at the three sites. We randomly selected two quadrats from the five quadrants of each plot for the 15-mm and 30-mm (equivalent to local moderate rainfall and heavy rainfall, respectively (Song et al., 2019) infiltration experiments. A square steel frame (width × length × height of 500 mm × 500 mm × 300 mm) was inserted 15 cm into the soil of each quadrat after the plants on the ground were harvested and the soil samples were collected. In total, 3.75 L and 7.5 L (representing infiltration amounts of 15 mm and 30 mm, respectively) of Brilliant Blue FCF solution (4 g L^{-1}) were evenly applied to each frame at an irrigation rate of 30 mm/h and allowed to infiltrate. Brilliant Blue is often used as a dye tracer because of its good compromise between fluidity and visibility (Weiler and Flühler, 2004; Bargués Tobella et al., 2014; Zhang et al., 2018a; Luo et al., 2019). After the dye was added, black plastic was placed on the frame surface to prevent the solution from evaporating. The plastic cloth and frame were removed 12 h after the dye tracer experiment. The vertical soil profile was excavated at 5 cm intervals to the maximum depth limited by the maximum dye penetration depth. A digital single-lens reflex (DSLR) camera (EOS 800D, Canon, Tokyo, Japan) consisting of a 6000 × 4000 pixel image sensor with an 18–35-mm lens (SIGMA 18–35 mm f/1.8 DC HSM ART) was used to obtain digitally stained images of each soil profile in full sun at noon. The focal length (18 mm), lens angle (parallel to the vertical soil profile), ISO (100), aperture (f/2.8), and shutter speed (1/200) of the lens did not change during the entire shooting process to minimize shooting errors.

The stained soil profile images were corrected using Adobe Photoshop CS5 (Zhang et al., 2018a) for geometric distortion, and then, a supervised classification method was used to convert the images into binary images with a resolution of 500 pixels × 500 pixels (Fig. 1). Root samples were collected in four soil layers (0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm) using a rectangular metal box (Wang et al., 2020) after the images were obtained. A total of 216 root samples (three sites × three plots × two infiltration amounts × four layers × three sampling replicates) were collected for laboratory analysis. The intact samples with soil and roots were washed on a 0.25-mm sieve. The roots were carefully extracted and then scanned at 1200 dpi using a 11000XL scanner (Epson, Tokyo, Japan) for the root parameter measurements. Root images were then

Table 1
Altitude, geographic position, and community properties under different types of grazing management sites.

Site	Altitude (m)	Geographic position	Dominant plant species	Plant height (cm)	Vegetation coverage (%)	R	H	E
GE	1603	N41°21′8″ E111°12′26″	*Stipa krylovii, Leymus chinensis, Agropyron cristatum*	25.40 ± 1.80a	92.10 ± 2.10a	7.20 ± 0.69a	1.86 ± 0.20a	0.95 ± 0.07a
CG	1599	N41°21′21″ E111°12′30″	*Stipa krylovii, Leymus chinensis, Caragana stenophylla*	18.35 ± 1.25b	60.07 ± 3.30b	6.13 ± 0.12b	1.62 ± 0.06ab	0.90 ± 0.02a
FG	1601	N41°21′18″ E111°12′34″	*Stipa krylovii, Artemisia frigida, Agropyron cristatum*	11.20 ± 1.71c	48.80 ± 3.05c	5.47 ± 0.23b	1.53 ± 0.09b	0.90 ± 0.03a

Note: Richness index (R), Shannon-Wiener diversity index (H), evenness index (E). The same letter means the differences are not significant at $p = 0.05$.

Table 2
Soil properties at the 0–40-cm soil depth in the grazing exclusion (GE), cold-season grazing (CG), and free-grazing (FG) grasslands.

Site	BD (g cm^3)	SWC (%)	TP (%)	SOC (g kg^{-1})	MWD (mm)	Particle size composition		
						Clay (%)	Silt (%)	Sand (%)
GE	1.43 ± 0.06a	13.01 ± 0.93a	46.69 ± 1.85a	22.67 ± 0.82a	3.00 ± 0.30a	5.32 ± 0.22a	20.15 ± 0.87a	74.50 ± 1.05a
CG	1.51 ± 0.04ab	8.87 ± 1.62b	44.27 ± 1.48ab	21.87 ± 1.7a	2.91 ± 0.62a	4.59 ± 0.28b	17.97 ± 1.64a	77.41 ± 1.93a
FG	1.60 ± 0.06b	7.57 ± 1.41b	41.07 ± 1.85b	18.54 ± 3.18a	1.69 ± 0.09b	3.56 ± 0.36c	12.43 ± 1.13b	83.98 ± 1.49b

Note: Values are means ± SD ($n = 9$). Bulk density (BD), soil water content (SWC), total porosity (TP), soil organic carbon (SOC), mean weight diameter (MWD). Different letters indicate significant differences ($p < 0.05$) among different sites.

analyzed using WinRHIZO Pro 2012b software (Regent Instruments Inc., Quebec City, Canada). Then, all roots were dried at 60 °C for 48 h to a constant weight to determine the belowground biomass (BGB) after the root parameter measurements were completed (Cui et al., 2019). The root morphological traits were described in terms of root length density (RLD), Diameter (mean root diameter), and specific root length (SRL) (Hao et al., 2020a).

2.4. Quantification of preferential flow

2.4.1. Characteristic indices of preferential flow

Preferential flow was quantified using the following indicators based on the analysis of the stained soil profile images (see Fig. 1 for visual examples).

Dye coverage (DC,%) represents the proportion of the stained area to the total soil profile (Flury et al., 1994):

$$DC = \left(\frac{D}{(D + ND)}\right) \times 100, \tag{7}$$

where DC denotes the percentage of the stained area; and D and ND denote the respective dye-tracing area and unstained area.

The preferential flow fraction (PF_{fr},%) represents the percentage of total infiltration through the pathways of preferential flow (Van Schaik, 2009) and was calculated as follows:

$$PF_{fr} = \left(1 - \frac{Unifr \times W}{TSA}\right) \times 100, \tag{8}$$

where PF_{fr} denotes the preferential flow fraction (%), Unifr denotes the matrix flow depth (cm), W denotes the soil profile width (50 cm in this study), and TSA represents the stained area of the total soil profile (cm^2).

The length index (LI) represents the absolute differences between the DC values and the vertical profile depth (Bargués Tobella et al., 2014):

$$LI = \sum_{i=1}^{n} |DC_{i+1} - DC_i|, \tag{9}$$

where LI denotes the length index, DC_i and DC_{i+1} represent the dye coverage in the i and i + 1 soil layer, and n denotes the number of pixels in the stained image (500 pixels in this study).

MaxD is the maximum depth of the stained area (cm).

2.4.2. Multi-index evaluation of preferential flow

We used a multi-index evaluation method to comprehensively evaluate the degree of preferential flow. First, a range method was used to standardize the preferential flow index $G_j = \{G_1, G_2, \ldots, G_6\}$($G_1, G_2, \ldots, G_6$ are TSA,

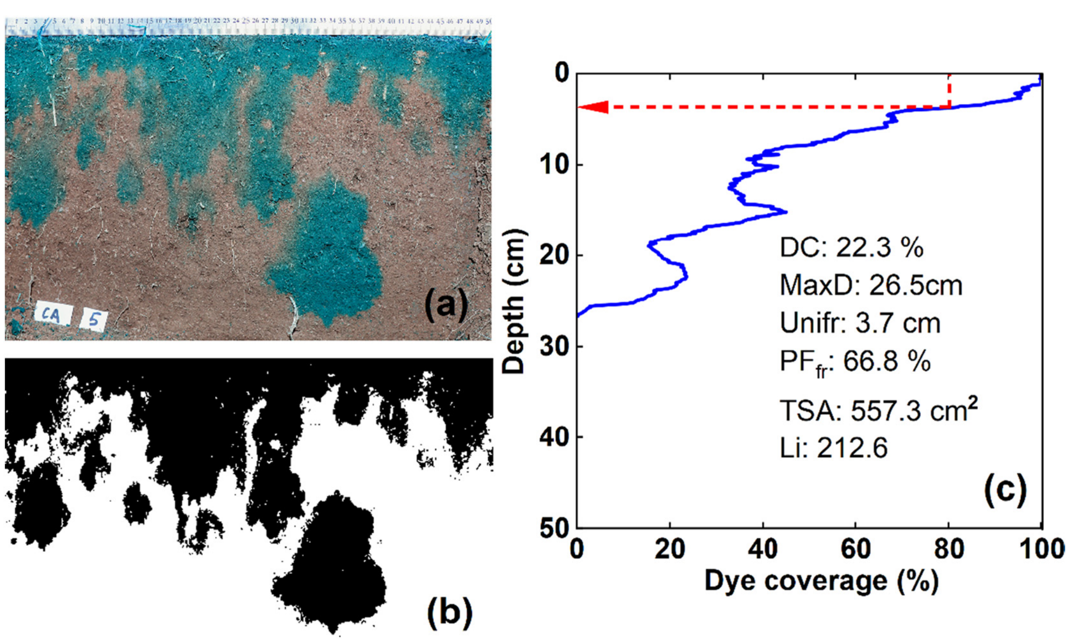

Fig. 1. Example of a vertical dye-tracer infiltration profile for the GE site and 30-mm infiltration events (a), corresponding binary digital image (b), and preferential flow parameter extraction (c). The blue curve represents the variation of the profile dye coverage with soil depth. The red dashed line represents the soil depth corresponding to the dye coverage = 80% (Unifr).

Unifr, MaxD, DC, PF_{fr}, and LI, respectively) of the different types of grazing management sites A = {GE, CG, FG} to obtain the standardized value Z_{ij}, and then, we calculated the mean square error $\sigma(G_j)$ of each index. Finally, the weight coefficient of each index was determined by the mean square error decision-making method. The calculation steps are as follows:

Relative calculation

Preferential flow index normalization:

$$Z_{ij} = \frac{\left|X_{ij} - X_{min}\right|}{X_{max} - X_{min}}, \tag{10}$$

where Z_{ij} is the standardized value of the index; X_{ij} is the measured value of the preferential flow index; X_{min} is the maximum value of the index; X_{max} is the minimum value of the index; i is the ordinal number of m data for a given index; and j is the number of indexes (j = 1, 2, …, 6).

The mean of the normalized index:

$$E(G_j) = \frac{1}{m}\sum_{i=1}^{m} Z_{ij}. \tag{11}$$

The mean square error of each index:

$$\sigma(G_j) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(Z_{ij} - E(G_j)\right)^2}. \tag{12}$$

The weight coefficient of each index:

$$W_j = \frac{\sigma(G_j)}{\sum_{j=1}^{6}\sigma(G_j)}. \tag{13}$$

The preferential flow evaluation index P_{FI} can directly reflect the degree of soil preferential flow:

$$P_{FI} = \sum_{i=1}^{m} Z_{ij}W_j. \tag{14}$$

2.4.3. Classification of preferential flow

Preferential flows were classified into five different types using the same image analysis and flow type classification as described in Weiler and Flühler (2004). This is based on the area of the stained path width (SPW) types. The SPW is equivalent to the horizontal range of the dye flow path (Hartmann et al., 2021).

2.5. Statistical analyses

The soil properties, vegetation properties, and root morphological traits were first tested by the Shapiro-Wilk method for normal distribution, and then, logarithmic or square root transformation was performed on non-normally distributed variables. Then, one-way analysis of variance (ANOVA) with Duncan's post hoc test was applied to assess the significant differences among soil properties and vegetation characteristics and the preferential flow indexes under the different types of grazing management ($p < 0.05$). Pearson correlation analyses were conducted to determine the relationships among the preferential flow index, soil properties, and vegetation characteristics. In addition, we performed path analysis to assess the direct and indirect effects of the soil and plant properties on the soil preferential flow. All of the statistical analyses were performed using SPSS 24.0 software (SPSS Inc., Chicago, Illinois, USA).

3. Results

3.1. Soil properties, root traits, and vegetation attributes

At the GE site, the BD was significantly lower than that at the FG site, and the TP was significantly higher than that at the FG site ($p < 0.05$) (Table 1). However, no significant differences were found for BD and TP between CG and GE sites or between CG and FG sites ($p > 0.05$). The SWC at the GE site was 13.01%, which was 4.14% and 5.44% higher ($p < 0.05$) than that at the CG and FG sites, respectively, whereas no significant differences were found between the CG and FG sites. There were no significant differences in MWD between the GE and FG sites ($p > 0.05$), but the MWD at the GE and CG sites was significantly higher than that at the FG site. The soil clay content differed markedly among the three sites; the highest average clay content was measured at the GE site (5.32%), followed by the CG (4.59%) and FG sites (3.56%). However, there were no significant differences in SOC among the three grazing management sites. The response patterns of the silt content and sand content to the grazing management were similar to that of the MWD (Table 1).

Root traits varied among the three sites (Fig. 2). The changes in the Diameter and RLD decreased rapidly with increasing soil depth. Conversely, the SRL increased with the soil depth (Fig. 2). In terms of the Diameter, the values at the GE site were significantly higher than those at the CG and FG sites ($p < 0.05$). There were no significant differences in Diameter between the CG and FG sites at the soil depth of 0–30 cm ($p > 0.05$). With respect to the SRL, significantly higher values were found at the CG and FG sites than at the GE site along the soil profile. Similarly, no significant differences were found in SRL between the CG and FG sites at the 0–40-cm soil depth ($p > 0.05$). Regarding RLD, no significant differences were found for the 0–20-cm soil depth between the GE and FG sites. However, the RLD at the CG site was significantly lower than that at the GE and FG sites at the soil depth of 0–20 cm ($p < 0.05$). Furthermore, significant differences in RMD were observed among the three sites at the soil depth of 0–20 cm ($p < 0.05$). In contrast, the RMD did not significantly change among the three sites at the 20–40-cm soil depth ($p > 0.05$; Fig. 2).

The vegetation attributes varied markedly among the three grazing management sites (Table 1). The respective plant height and vegetation coverage at the GE site were 25.40 cm and 92.10%, which were respectively 7.05 cm and 32.03% higher than those at the CG site and were respectively 14.20 cm and 43.3% higher than those at the FG site. The richness index (R) and Shannon-Wiener diversity index (H) values of the GE site were significantly higher than those of the FG site ($p < 0.05$). However, there were no significant differences in the evenness index (E) among the three grazing management sites ($p > 0.05$).

3.2. Preferential flow behavior and infiltration patterns

Three-dimensional visualizations of soil infiltration at the GE, CG, and FG sites showed that spatial distributions of water infiltration were distinctly different among the three grazing management practices (Fig. 3). At an infiltration amount of 15 mm, as indicated by the red and light green areas, a higher infiltration depth generally appeared at the GE site. The information of the stained profile image directly reflects the infiltration behavior of the water flow (Table 3). The mean MaxD and DC of the GE, CG, and FG sites varied from 6.93 to 10.27 cm and from 8.55 to 11.76%, respectively. The MaxD and DC values at the GE site were significantly higher than those at the CG and FG sites ($p < 0.05$), indicating that a larger amount of water preferentially infiltrated into deeper soil layers at the GE site than at the CG and FG sites (Fig. 3). The mean P_{FI} of the GE, CG, and FG sites was 0.89, 0.30, and 0.15, respectively (Table 4), which was consistent with the three-dimensional images of soil infiltration (Fig. 3a–c). Moreover, the mean P_{FI} of the GE site was 2.97 and 5.93 times greater than that of the CG and FG sites, respectively. A higher P_{FI} represents greater preferential flow infiltration with soil depth. At an infiltration amount of 30 mm, we observed mainly preferential flow with deep infiltration at both the GE and CG sites (Fig. 3d–f). There were no significant differences in MaxD between

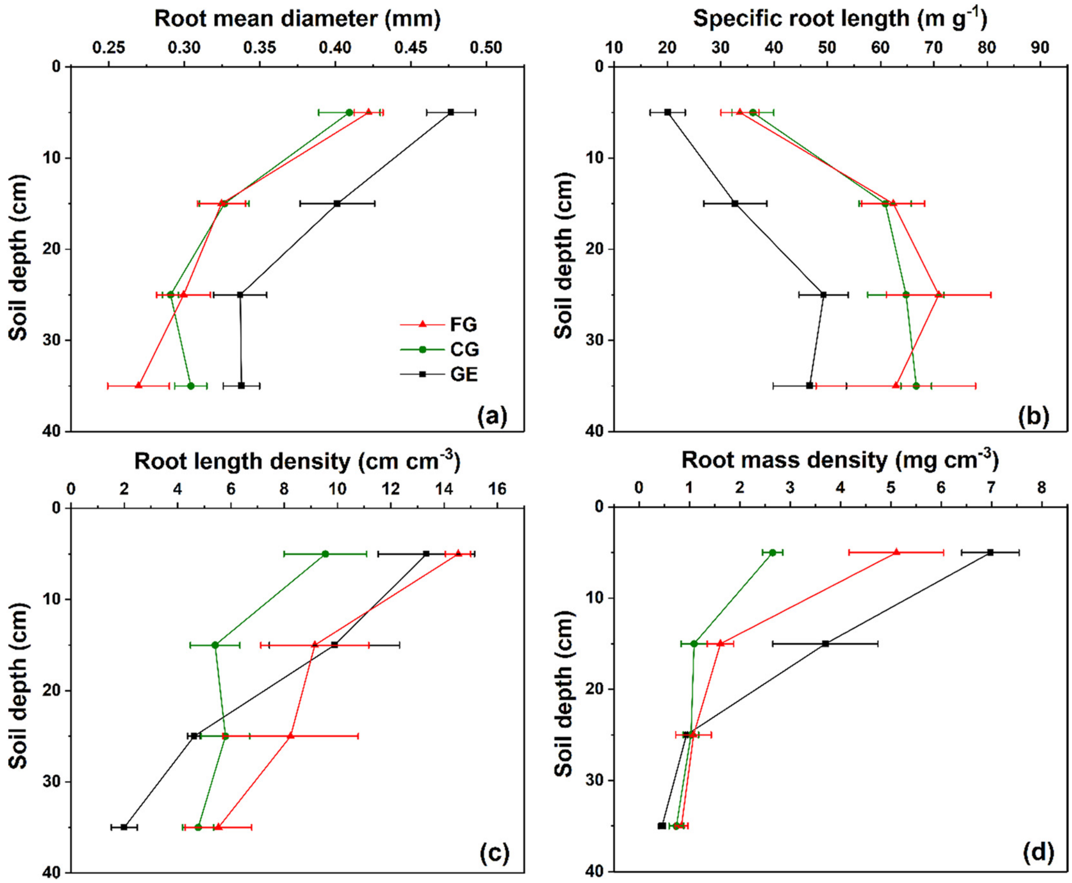

Fig. 2. Changes in the root morphological traits with soil depth between the different types of grazing management.

the GE and CG sites. However, the mean MaxD of the GE and CG sites was significantly greater than that of the FG site by 54.17% and 43.66%, respectively. The mean P_{FI} of the GE, CG, and FG sites was 0.89, 0.62, and 0.17, respectively. Similarly, the DC differed markedly among the three sites.

Compared with 15-mm infiltration, as indicated by the red and light green areas, a deeper infiltration depth occurred under 30-mm infiltration. The Unifr, MaxD, and DC values of the soil profile for the 30-mm infiltration amount at the three sites were significantly greater than those for the 15-mm infiltration amount (Table 3). The LI significantly differed only between the 15-mm and 30-mm infiltration amounts at the CG site. A higher LI represents greater heterogeneity of the macropore flow pattern. Moreover, the mean P_{FI} of the CG site was significantly improved under 30-mm infiltration. This indicates that the infiltration pathways at the CG site differed significantly between the 30-mm and 15-mm infiltration amounts.

Different SPW distribution patterns along the soil profiles were observed among the three experimental sites (Fig. 4). In the upper soil layer (<25 mm depth in Fig. 3a, <50 mm depth in Fig. 3b), >200 mm width staining paths dominated, followed by 20–200 mm and <20 mm staining widths. The proportion of 20–200 mm and <20 mm width staining paths gradually increased with the increase in soil depth and was frequently dominant in the subsoil (>75 mm depth).

Consistent with the SPW distribution, the distribution of preferential flow types also showed significant differences between different sites and different soil depths (Fig. 5). Homogeneous matrix flow was mostly observed in the topsoil layers, which accounted for 15.01%, 19.20%, and 22.11% at the GE, CG, and FG sites, respectively, for the 15-mm infiltration amount (Fig. 4a). Inversely, macropore flow, especially with mixed interaction and low interaction, dominated the preferential flow in the subsoil (>40 mm depth), accounting for 61.88%, 59.96%, and 49.83% at the GE, CG, and FG sites, respectively. Moreover, with the increase in the infiltration amount to 30 mm, the proportion of macropore flow with high interaction and mixed interaction increased, accounting for 44.47%, 47.47%, and 48.44% at the GE, CG, and FG sites, respectively (Fig. 5b). The proportion of macropore flow with high interaction under the 30-mm infiltration amount was 91.6%, 159.6%, and 6.0% higher than that of the 15-mm infiltration amount for the GE, CG, and FG sites, respectively.

3.3. Factors influencing preferential flow

The Pearson correlation tests confirmed that preferential flow had a significantly positive correlation with plant coverage, plant height, MWD, and SWC ($p < 0.01$), clay content and silt content, ($p < 0.001$), and R, Diameter, TP, and SOC ($p < 0.05$), whereas it was strongly and negatively correlated with the sand content ($p < 0.001$) and SRL and BD ($p < 0.05$) (Fig. 6). Moreover, the H, E, and RLD had no significant correlation with soil preferential flow ($p > 0.05$). The preferential flow was significantly affected by multiple factors according to the Pearson correlation analysis. The path analysis indicated that preferential flow was mainly affected by the clay content, E, diameter, and TP (Table 5). The clay content, diameter, and TP had a direct positive effect on preferential flow, with path coefficients of 0.53, 0.40, and 0.45, respectively. E had a direct negative effect on preferential water flow, with a path coefficient of −0.64. Clay, diameter, and TP all had indirect positive effects on preferential flow, whereas E had an indirect negative effect on preferential flow. Noticeably, the indirect effects of clay, diameter, and TP were lower than their direct effects (Table 5). Conversely, the direct effects of E were lower than its indirect effects. As mentioned above, clay, E, diameter, and TP all had positive effects on soil preferential flow, with path coefficients of 0.92, 0.17, 0.68, and 0.77, respectively (Table 5).

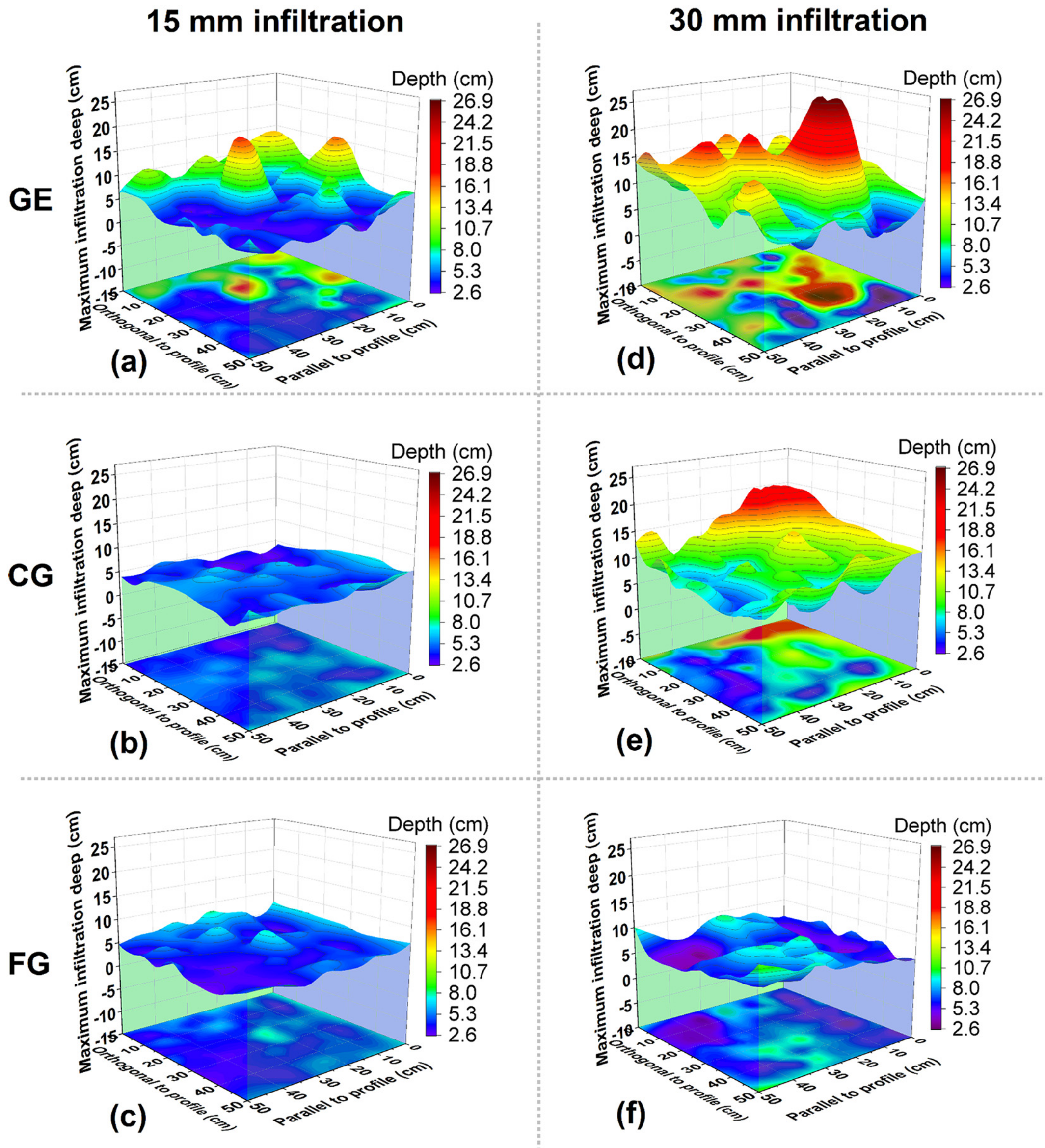

Fig. 3. Spatial distribution of preferential flow under different types of grazing management and infiltration amount. (a), (b), and (c) correspond to the grazing exclusion (GE), cold-season grazing (CG), and free-grazing (FG) grassland sites under the 15-mm infiltration amount. Similarly, (d), (e), and (f) correspond to the three sites (GE, CG, and FG) under the 30-mm infiltration amount. The ground surface of each plot area was discretized into a square grid of 5 cm × 5 cm, thereby obtaining 100 square grids for each dye-tracer experiment. The 3D maps of the preferential flow spatial distribution were produced with Origin Pro 2021b (OriginLab Corporation, MA, USA).

4. Discussion

4.1. The effect of grazing management on soil and vegetation properties

Land use can cause significant changes in the evolution of the physical properties of the soil (Sun et al., 2018a; Zhang et al., 2021). The soil properties (i.e., BD, TP, MWD, and the soil particle size composition) of the three sites (GE, CG, and FG) were greatly affected by the grazing management practices (Table 2). The mismanagement of excessive numbers of animals could lead to the compaction of grassland soils (Pulido et al., 2017). Under FG conditions, trampling by livestock causes soil deformation through the frequent application of ground contact pressure beneath their hooves (Warren et al., 1986). The soil compaction process is mainly related to a decrease in soil volume, which leads to an increase in soil BD (Alaoui et al., 2011), changes in pore connectivity (Zhao et al., 2010), and an increase in the resistance of root penetration (Sun et al., 2018b). Conversely, livestock was completely excluded from the GE site, which could directly inhibit overgrazing and hoof trampling, thereby reducing soil compaction and BD (Table 2). The shifts in particle size composition under grazing management can be explained by the changes in plant coverage, plant height, and soil erosion (Teague et al., 2011; Jiang et al., 2021). This was confirmed by Du et al. (2019), who found that the soil particles from the CK plot to the HG plot showed a coarsening trend by the sorting action of wind.

How grazing management impacts community composition is a key question in ecological research (Yu et al., 2020). In the present study, we found that plant height, vegetation coverage, R, and H of the GE site were significantly higher than those of the CG and FG sites, which is consistent with the results of previous studies on Inner Mongolian grasslands (Wang et al., 2017; Liu et al., 2018). The species composition under grazing management is determined by the differences in plant tolerance and herbivore selectivity (Yu et al., 2020). However, long-term overgrazing can also lead to decreased selectivity of herbivores (Wan et al., 2015). Therefore, the impact of grazing management on community composition may be closely related to plant tolerance. The optimal partitioning hypothesis states that plants reduce the proportion of AGB and allocate more photosynthetic products to BGB in response to environmental stress (Sun et al., 2021). In

Table 3
Statistical summary of the preferential flow parameters from parallel cross-sectional images in the grazing exclusion (GE), cold-season grazing (CG), and free-grazing (FG) grasslands with three replicates.

Infiltration amount (mm)	Site	Plot	Unifr (cm)	MaxD(cm)	DC (%)	PF_{fr}(cm)	LI (%)
15	GE	1	2.95	11.42	11.16	46.13	134.88
		2	3.93	9.28	11.30	30.41	110.71
		3	4.30	10.12	12.80	32.53	119.52
		Mean	3.73 ± 0.70aA	10.27 ± 1.08aA	11.76 ± 0.91aA	36.36 ± 8.53aA	121.7 ± 12.23aA
	CG	1	3.48	8.20	9.86	28.92	122.68
		2	3.49	6.55	9.22	24.56	112.50
		3	3.33	6.12	8.50	21.64	106.37
		Mean	3.43 ± 0.09aA	6.96 ± 1.10aB	9.19 ± 0.68aB	25.04 ± 3.67aA	113.85 ± 8.24aA
	FG	1	3.29	6.49	8.57	22.90	105.19
		2	2.78	7.93	8.44	33.75	110.58
		3	2.82	6.38	8.65	26.78	106.56
		Mean	2.96 ± 0.28aA	6.93 ± 0.87aB	8.55 ± 0.11aB	27.81 ± 5.50aA	107.44 ± 2.80aA
30	GE	1	5.00	19.07	21.35	53.77	162.75
		2	7.17	15.44	20.82	30.71	128.62
		3	7.99	16.56	22.41	28.64	143.03
		Mean	6.72 ± 1.54bA	17.02 ± 1.86bA	21.53 ± 0.81bA	37.71 ± 13.95aA	144.8 ± 17.13aA
	CG	1	6.57	15.90	19.17	31.43	132.14
		2	7.70	14.70	20.36	26.77	131.91
		3	4.70	16.99	15.95	39.23	132.01
		Mean	6.32 ± 1.51bA	15.86 ± 1.15bA	18.49 ± 2.28bB	32.48 ± 6.30aA	132.02 ± 0.11bAB
	FG	1	4.37	11.69	14.37	38.98	123.51
		2	5.38	12.03	15.07	28.66	109.32
		3	5.01	9.41	13.31	25.02	111.16
		Mean	4.92 ± 0.51bA	11.04 ± 1.42bB	14.25 ± 0.89bC	30.89 ± 7.24aA	114.66 ± 7.72aB

Note: Values are means ± SD ($n = 3$). Uniform infiltration depth (Unifr), maximum depth of the stained area (MaxD), dye-tracing area ratio (DC), preferential flow fraction (PF_{fr}), length index (LI). Different lowercase letters indicate significant differences ($p < 0.05$) among infiltration amounts within a site, and different uppercase letters indicate significant differences ($p < 0.05$) among different sites under the same infiltration amount.

our study, we observed a significant decrease in SWC, TP, and clay content but increased BD from the GE site to the FG site (Table 2), which indicates a deterioration of the soil environment after disturbance by grazing (Armitage et al., 2012). Therefore, plants must allocate more biomass to the root system in order to more effectively absorb water and nutrients under FG conditions (Fig. 2d). Plants at the FG site increased their SRL by increasing their root length to improve their competitiveness (Fig. 2b). However, the SRL at the CG and FG sites remained similar in a grazing environment, but the architectural strategy differed. The FG site was grazed year-round, but the CG site was completely excluded during the growing season. Accordingly, the plants at the CG site needed to allocate more photosynthetic products aboveground in order to increase their competitive advantage during the growing season (Sun et al., 2021), resulting in a decrease in BGB (Fig. 2d). Conversely, livestock were completely excluded from the GE site, and thus, the soil bulk density is relatively small (Table 2) and less resistant to root growth, which is more conducive to BGB accumulation (Fig. 2d). Moreover, with the decrease in water and nutrient absorption demand pressure, plants may use more BGB for the construction of root system integrity (i.e., root diameter) (Zamora et al., 2007), resulting in a decrease in SRL (Fig. 2b). Furthermore, we found that the root traits, namely, RMD, SRL, and diameter, were correlated with R and H (Fig. 6). This implied that the differences in community composition under grazing management can be explained by the changes in SRL and diameter related to root traits.

Table 4
Evaluation index (P_{FI}) of soil preferential flow at different sites.

Infiltration amount (mm)	Site	Evaluation index of preferential flow			
		Min value	Max value	Standard deviation	Mean value
15	GE	0.76	1.04	0.14	0.89
	CG	0.10	0.54	0.22	0.30
	FG	0.09	0.2	0.05	0.15
30	GE	0.76	0.98	0.12	0.89
	CG	0.46	0.74	0.14	0.62
	FG	0.05	0.23	0.10	0.17

The aggregates at the GE and CG sites had a significantly higher MWD than those at the FG site. However, there was no significant difference in the MWD between the GE and CG sites. These results indicate that grazing management significantly affected the stability of soil aggregates. Under FG conditions, the trampling of animals caused soil deformation, and then, the aggregates were broken down gradually as the trampling time increased, resulting in a decrease in the soil macrospore volume (Yi et al., 2020), which also partly explained the decrease in TP from the GE site to the FG site. Nevertheless, plant roots also play an important role in soil aggregate stability at a local scale (Hao et al., 2020a; Hao et al., 2020b). Root morphological traits include root length and diameter, which directly affect soil stability by binding and compressing soil particles (Garcia et al., 2019; Hao et al., 2020b). However, there was no significant relationship between the root traits and MWD, but there were significant relationships between root traits and coverage, height, R, and H in our study, suggesting that the root traits did not directly affect the aggregate stabilities. Although plant roots play an important role in modulating various soil properties (Garcia et al., 2019), the frequent ground contact pressure beneath the hooves may have overridden the positive effect of plant roots on the soil physical properties. Moreover, frequent trampling would also cause obstacles to root growth (Zhao et al., 2010). Taken together, the shifts in soil and vegetation properties under grazing management can be explained by the changes in mechanical stress (e.g., livestock trampling) and plant tolerance (e.g., root traits and community composition).

4.2. The effects of soil and vegetation properties on preferential flow behavior and infiltration patterns

Preferential water flow behavior elucidated by three-dimensional visualizations was complicated and irregular (Fig. 3). Obviously different types of preferential flow were observed at the three sites. Soil from the GE site had greater preferential flow compared with the CG and FG sites under 15-mm infiltration. Many previous studies indicated that the preferential flow behavior was mainly affected by the changes in the soil physical properties and hydrological conditions (Bargués Tobella et al., 2014; Zhang et al., 2015; Wang et al., 2018; Yi et al., 2020). In the present study, the GE site was totally excluded from the FG site, which could directly inhibit

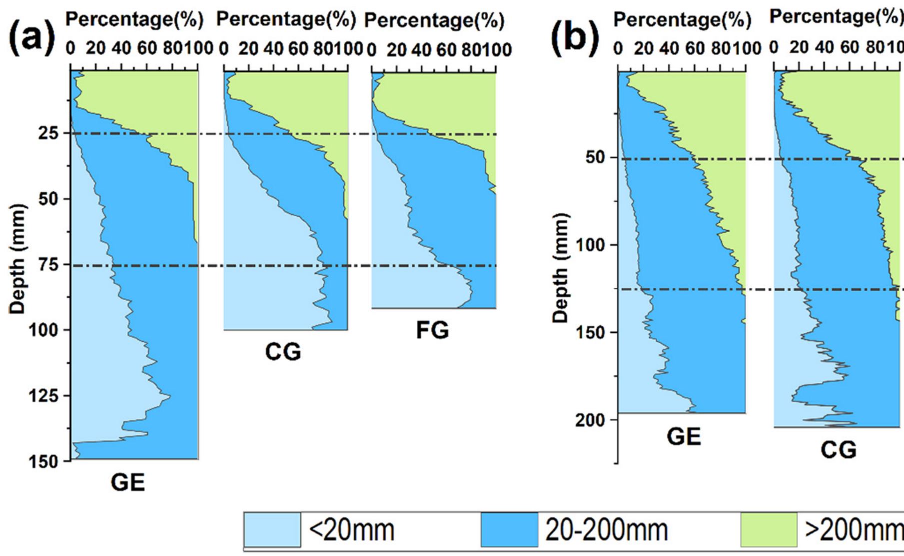

Fig. 4. The average stained area ratio of three stained path width classes at the grazing exclusion (GE), cold-season grazing (CG), and free-grazing (FG) grassland sites under the 15-mm (a) and 30-mm (b) infiltration amounts.

overgrazing and hoof trampling, thereby reducing soil compaction and BD (Table 2). Soil BD, which is negatively related to TP, MWD, SWC, and P_{FI}, is a key soil physical factor that influenced the preferential flow behavior in our study (Fig. 6). Correlation analysis indicated that preferential flow was closely related to plant coverage, height, and richness (Fig. 6). These results are consistent with previously reported results (Fischer et al., 2015; Liu et al., 2019; Zhu et al., 2020). In addition, a previous study suggested that in macroporous soils, higher initial soil moisture conditions increase the depth of macropore flow penetration because a higher initial soil moisture content reduces lateral flow into the soil matrix (Zhang et al., 2016). Sanders et al. (2012) also pointed out that dry initial conditions will lead to a higher soil matrix sorption capacity. GE resulted in considerable canopy cover (Zhao et al., 2011), which can effectively intercept rainfall and control soil erosion by water and wind (Jiang et al., 2021), thus subsequently preventing soil consolidation (Sun et al., 2018b) and thereby increasing soil infiltration and the SWC (Wang et al., 2019; Liu et al., 2020a). The soil particle size distribution is often related to infiltration properties, which have a great influence on hydraulic characteristics (Tang et al., 2019). Previous studies suggested that the sand content had a positive effect on water infiltration (Chartier et al., 2011; Mei et al., 2018). However, we found strong positive direct and indirect effects of the clay content on preferential flow according to path analysis (Table 5). This difference may be due to the different soil textures at our research site compared with the sites investigated by Mei et al. (2018) and Chartier et al. (2011).

Soil physical properties, hydrological properties, and water flow behavior are closely related to root morphology (Bargués Tobella et al., 2014; Liu et al., 2020b). Plant root systems can create powerful new flow paths (Hu et al., 2019), which can establish connectivity between existing preferential flow paths and make them more hydrologically active (Jiang et al., 2018; Yi

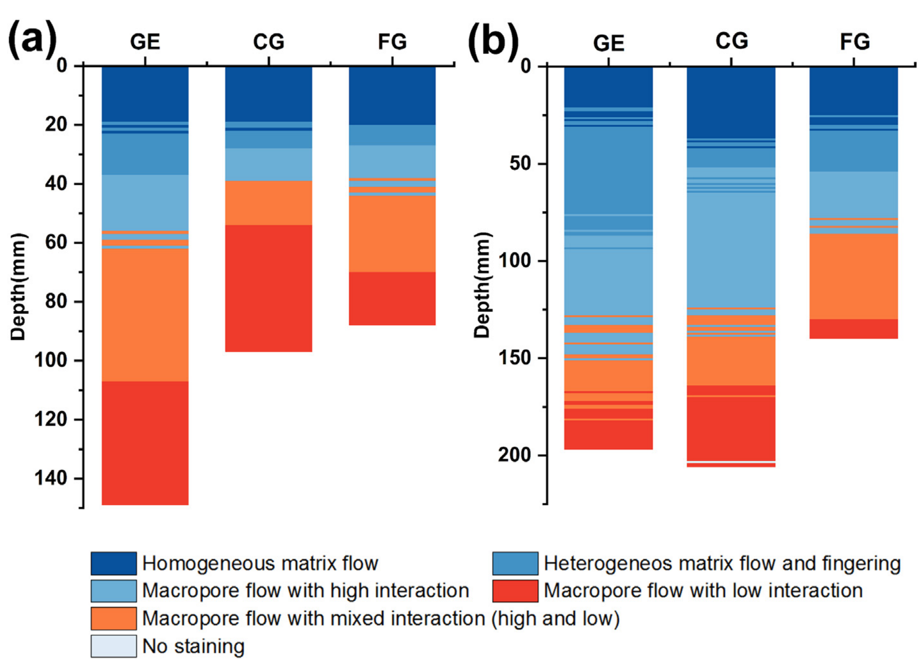

Fig. 5. Preferential flow types (classified based on stained path width distribution) at the grazing exclusion (GE), cold-season grazing (CG), and free-grazing (FG) grassland sites under the 15-mm (a) and 30-mm (b) infiltration amounts.

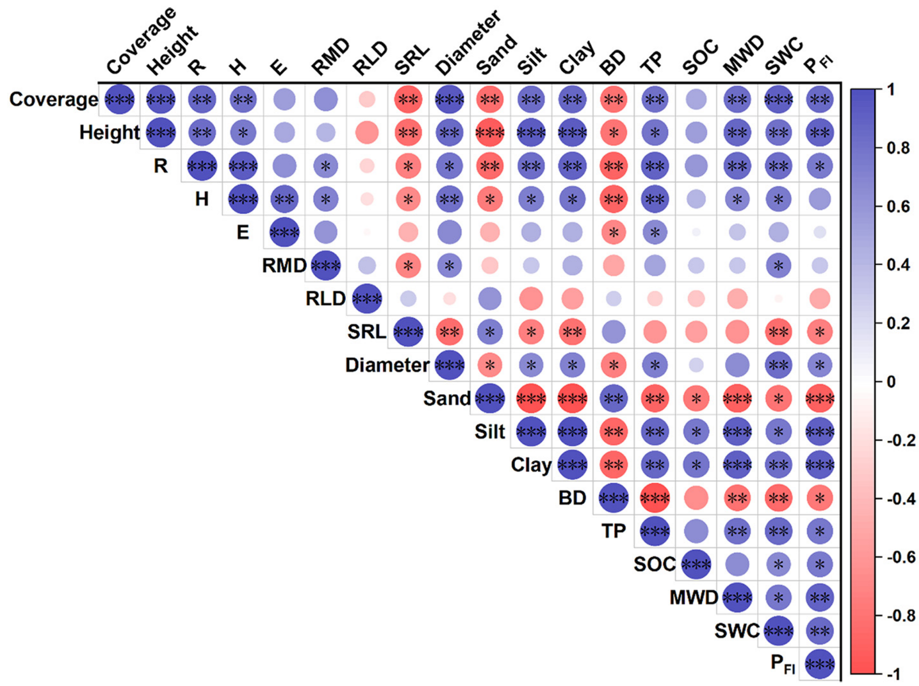

Fig. 6. Pearson correlation analysis of the main vegetation and soil physical properties. Coverage, the average plant coverage within each quadrat; Height, the average height of each quadrat; R, Richness index, H; Shannon-Wiener diversity index; E, Evenness index; RMD, root mass density; RLD, root length density; SRL, specific root length; Diameter, root mean diameter; Sand, sand content (%); Silt, silt content (%); Clay, clay content (%); BD, bulk density; TP, total porosity; SOC, soil organic carbon; MWD, mean weight diameter; SWC, soil water content; P_{FI}, the preferential flow evaluation index. Indicated values represent the correlation coefficients. The blue color indicates a positive correlation, and the red color indicates a negative correlation at significance levels of *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

et al., 2020). Plant roots under GE promote the formation of soil macropores, thereby enhancing total porosity (Table 2), which allows preferential water infiltration into deep soil layers (Fig. 2). Therefore, greater preferential flow and higher infiltration heterogeneity at the GE site could be closely related to the well-developed root systems at this site (Zhao et al., 2011; Hu et al., 2019).

The water exchange between the macropores and the surrounding soil matrix is an important process of water infiltration (Jiang et al., 2015). The initiation vertical macropore flow near the soil surface caused the largest water exchange between the macropores and the soil matrix (Zhang et al., 2018a) (Fig. 5), and the matrix flow was the dominant flow type in the 0–5-cm soil layer (Yi et al., 2020). However, the water exchange with the soil matrix decreased with the soil depth (Fig. 5). The factors that influence the interaction between the macropores and the surrounding soil matrix were different among the three grazing sites (Fig. 5). Long-term GE promotes the natural restoration of total pores and forms pores dominated by plant roots (Dong et al., 2015; Hu et al., 2018), resulting in high permeability and good vertical connectivity along the soil profile (Jiang et al., 2018). The soil hydrological properties were affected by grazing management (Table 2); higher SWC and TP at the GE site together caused the preferential infiltration to bypass the soil matrix and consequently resulted in low interaction macropore flow in the subsoil (Jiang et al., 2015; Mei et al., 2018) (Fig. 5). Moreover, the unbroken pores facilitated the lateral infiltration of water from the macropores to the soil matrix, and the lower initial SWC of the CG and FG sites (Table 2) together contributed to higher interaction (Jiang et al., 2015). Although the CG site had a greater depth of infiltration under 30-mm infiltration, higher interaction will slow down the downward movement of water in the macropores (Cey and Rudolph, 2009), which allows more water to remain on the surface of the soil, increasing the risk of soil evaporation. We observed that the proportion of macropore flow with high interaction under the 30-mm infiltration amount was 91.6%, 159.6%, and 6.0% higher than that under the 15-mm infiltration amount for the GE, CG, and FG sites, respectively (Fig. 5). This indicated that a high infiltration amount (30 mm) can improve interactions between macropores and the soil matrix (Jiang et al., 2015). Weiler and Flühler (2004) similarly found that the irrigation rate affects the interactions between macropore flow and soils, with a higher irrigation rate resulting in macropore flow with a higher interaction. Therefore, the present results further show that the water exchange between the macropores and the surrounding soil matrix is mainly affected by the properties of the pore systems, the SWC, and the infiltration amount.

Table 5
Path analysis results of factors affecting soil preferential flow.

Item	Direct path coefficient	Indirect path coefficient				Total	Sum of coefficient
		Clay	E	Diameter	TP		
Clay	0.53		−0.29	0.28	0.39	0.38	0.92
E	−0.64	0.24		0.26	0.30	0.81	0.17
Diameter	0.40	0.38	−0.42		0.33	0.29	0.68
TP	0.45	0.47	−0.43	0.29		0.32	0.77

Note: Clay content (Clay), Evenness index (E), root mean diameter (Diameter), total porosity (TP).

4.3. Implications and limitations of the present study

Water shortage is the main factor restricting the construction and maintenance of regional ecosystems and the development of agricultural production in arid and semi-arid areas (Liu et al., 2020b). Understanding the direction and magnitude of the effects of different types of grazing management on the plant, soil, and preferential flow characteristics can greatly improve grassland management strategies (Tian et al., 2021). The present study confirmed the important role of plant root systems in preferential flow under grazing management. These effects may not be direct but may be mediated through soil physical properties (i.e., SWC, aggregate stability, and soil porosity). GE directly inhibited hoof trampling and excessive herbivory and enhanced plant density and root biomass, which could promote the natural restoration of total pores and preferential water infiltration. Hence, the greater infiltration and preferential water flow of GE could effectively increase preferential water and nutrient transport, which plays a vital role in the replenishment of deep soil water and the maintenance of vegetation in arid and semi-arid areas. Conversely, long-term FG increased BD and decreased plant coverage, height, and plant diversity, resulting in lower preferential flow, thus aggravating water shortage during plant growing seasons, especially in semi-arid areas where rainfall is limited. Therefore, long-term FG is not recommended as a sustainable grazing management practice. Moreover, since CG had lower potential for the recovery of soil water storage, our results suggested that GE could be a better grassland management practice than CG in the semi-arid grasslands of northern China. Furthermore, many arid and semi-arid regions worldwide have similar problems as our research sites. Therefore, the results of this study are not limited to the semi-arid grasslands in northern China.

The present study mainly focused on the effect of grazing management on preferential water flow. However, the response of preferential flow to grazing is affected not only by grazing management but also by different grazing intensities (Zhao et al., 2011) and grazing durations (Fanselow et al., 2011). Previous studies also indicated that grazing intensity had a significant effect on the accumulation of litter in production units ($p < 0.001$), and the reduction from GI0 to GI6 decreased by 83%, and the soil coverage decreased by 43% ($p < 0.001$), resulting in an increase in soil erosion vulnerability (Schönbach et al., 2011). In addition, we used three unique long-term demonstration sites to demonstrate the effects of grazing management on preferential water flow. Similar to experiments along chronosequences, this long-term grazing management experiment was inevitably a pseudoreplicated experiment, but within this constraint, we have done our best to follow the methods used in chronosequences and long-term pseudoreplicated experimental studies (Hafner et al., 2012; Guilherme Pereira et al., 2019; Yu et al., 2020).

5. Conclusions

This study investigated the effect of grazing management on soil properties, vegetation attributes, root traits, and preferential flow behavior. The GE site was excluded from the CG and FG sites, which could directly inhibit hoof trampling and excessive herbivory, thereby enhancing plant density and root biomass, which could promote the natural restoration of total pores and preferential water infiltration. The differences in community composition under grazing management can be explained by the changes in the SRL and root mean diameter related to root traits. The water exchange between the macropores and the surrounding soil matrix is mainly affected by the properties of the pore systems, SWC, and the infiltration amount. The present study highlights the connection between plant root systems and preferential flow under grazing management and that these effects may not be direct but mediated through soil physical properties (i.e., the SWC, aggregate stability, and soil porosity). The clay content, evenness index, root mean diameter, and total porosity were the main factors that influenced the preferential flow properties. The greater infiltration and preferential water flow under GE could effectively increase preferential water and nutrient transport, which plays a vital role in the replenishment of deep soil water and the maintenance of vegetation in arid and semi-arid areas.

CRediT authorship contribution statement

Xiaolong Wu: Conceptualization, Methodology, Writing – original draft. **Xiaohong Dang:** Methodology, Writing – review & editing. **Zhongju Meng:** Conceptualization, Supervision, Project administration. **Dongsheng Fu:** Investigation, Data curation. **Wencheng Cong:** Software, Investigation. **Feiyan Zhao:** Visualization. **Jingjie Guo:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (42067015), and the Natural Science Foundation of Inner Mongolia Autonomous Region (2020MS03038).

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